

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment considerations.

1. System Requirements

System Name: LibraryManagementSystem

Architecture Style: Microservices Architecture

Functional Requirements

- User registration and authentication
- Manage library members
- Catalog books and authors
- Search and browse books
- Borrow and return books
- Track borrowing history
- Send due date notifications

Non-Functional Requirements

- Scalability: Medium
- Availability: High
- Latency: Medium
- Reliability: High
- Consistency: Strong
- Security: Medium

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, connectors, and properties.

```
System LibraryManagementSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices Architecture";
  Property source : String = "AI";
  Component FaceDetectionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Detect the user's face using the YOLOv5 model, contours detection";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component GestureRecognitionModule = new Component {
    Port in;
```

```

    Port out;
    Property responsibility : String = "Recognize gestures using threading for concurrent execution";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component HandTrackingModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Track the user's hand using Background Subtraction (BS) to s
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component InteractionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Support single-hand interaction, exclude multiple-hand inter
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component Real-timeCollaborationModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Enable real-time collaboration by supporting interactions on
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component LatencyReductionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Analyze gestures and reflect the corresponding action on the
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment FaceDetectionModule.out to HandTrackingModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment HandTrackingModule.out to GestureRecognitionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment GestureRecognitionModule.out to InteractionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment InteractionModule.out to Real-timeCollaborationModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment Real-timeCollaborationModule.out to LatencyReductionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}

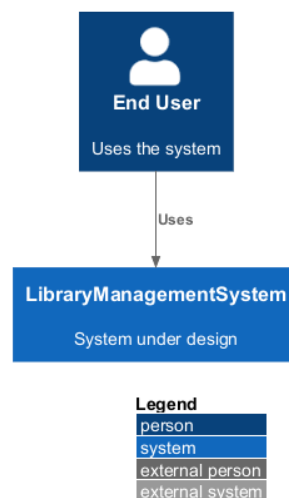
```

```

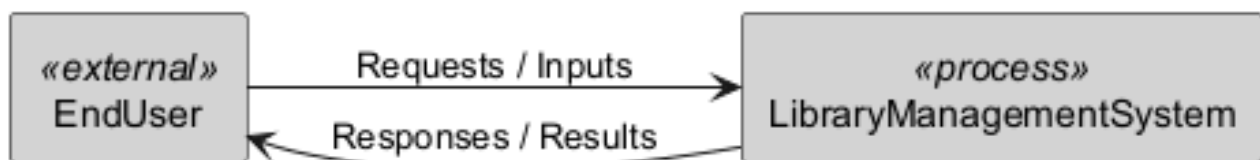
Attachment LatencyReductionModule.out to Real-timeCollaborationModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

3. Context View



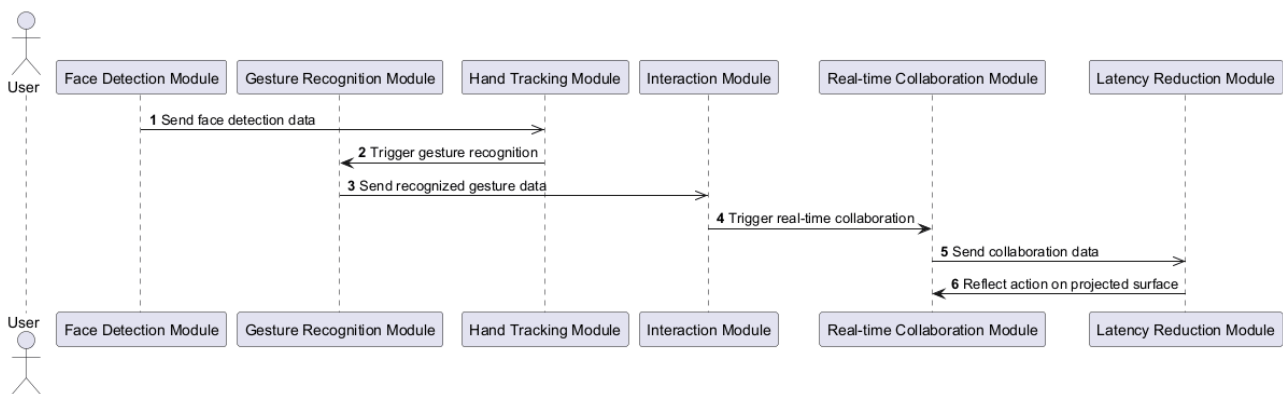
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

